

Advanced Web Design and Development – BIT3208 Practical Tasks

Weekly practicals for **BIT3208: Advanced Web Design and Development**, Bachelor of Business Information Technology, Mount Kenya University.

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Unit	BIT3208 – Advanced Web Design and Development
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This repository contains eight weeks of practical work, structured around one cumulative project – **ShopMart**, a PHP/MySQL e-commerce platform – plus three standalone practical tasks (Weeks 6–8) that each isolate a specific topic: database CRUD, session-based authentication, and responsive design.

Tech Stack

- **Backend:** PHP (mysqli extension, prepared statements throughout)
- **Database:** MySQL / MariaDB
- **Frontend:** HTML5, CSS3, vanilla JavaScript
- **Layout:** Flexbox, CSS Grid, mobile-first media queries
- **Local environment:** XAMPP (Apache, MySQL, PHP)
- **Tooling:** VS Code, phpMyAdmin, Postman, Chrome DevTools

No frameworks are used anywhere in this repository — that's deliberate. The brief asks for raw HTML/CSS/JS and PHP/mysqli so that every layer of the stack (routing, validation, queries, sessions) is written and understood directly rather than abstracted behind a framework.

Repository Structure

```
Advanced-web-design-Tasks/  
├── shopmart-week1/           # Week 1 – Local  
environment setup  
│   ├── hello.php  
│   ├── db_test.php  
│   └── http_test.php  
├── shopmart-week2/           # Week 2 –  
Wireframes & GUI design  
│   ├── wireframes.html  
│   └── gui-mockup.html  
├── shopmart-week3/           # Week 3 –  
JS/PHP basics  
│   └── shopmart-week3-validation.html
```

```

└─ shopmart-week4/                    # Week 4 – User
authentication
|   └─ register.php
|   └─ login.php
|   └─ dashboard.php
|   └─ logout.php
└─ shopmart-week5/                    # Week 5 – Full
CRUD (products)
|   └─ index.php
|   └─ create.php
|   └─ edit.php
|   └─ delete.php
|   └─ db.php
|
└─ week6-database-crud/                # Week 6 –
Database integration & CRUD
|   └─ class-activities/                # studentdb in-
class activities 1-3
|   |   └─ activity1_register.php
|   |   └─ activity2_display.php
|   |   └─ activity3_edit_delete.php
|   └─ employee-records-system/         # Challenge
Task – full system
|       └─ login.php
|       └─ index.php
|       └─ add_employee.php
|       └─ edit_employee.php
|       └─ delete_employee.php
|       └─ logout.php
|       └─ includes/auth.php
|
└─ week7-student-portal/              # Week 7 – Auth
& session management
|   └─ index.html
|   └─ register.php
|   └─ login.php
|   └─ dashboard.php
|   └─ logout.php

```

```
|
| └─ week8-responsive-design/      # Week 8 –
Responsive / mobile-first
|   └─ task1-profile-page/
|   └─ task2-product-showcase/
|   └─ crumb-and-co/              # Extra
assignment – 5-page site
|   └─ index.html
|   └─ about.html
|   └─ products.html
|   └─ training.html
|   └─ contact.html
|
└─ Documentation/
    └─ Consolidated_Logbook_Week1-8.docx
```

Week-by-Week Summary

Weeks 1–5 – ShopMart (cumulative project)

ShopMart is a KES-priced e-commerce platform built incrementally, with each week adding a layer on top of the last.

Week	Focus	What it adds
1	Local environment setup	Verifies the Apache/PHP/MySQL stack independently with three diagnostic scripts before any application code is written.

Week	Focus	What it adds
2	Wireframes & GUI design	Low-fidelity wireframes for the four core screens, followed by a high-fidelity dark-charcoal/emerald GUI concept.
3	JavaScript & PHP basics	Client-side registration form with live DOM-based validation (username, email format, password strength, confirm-password match).
4	User authentication	PHP + MySQL login system: bcrypt password hashing (password_hash / password_verify), prepared statements, native PHP sessions, protected dashboard.
5	Database components	Full CRUD product management module, reusing the Week 4 session guard so only authenticated users can reach it.

Planned next step: Safaricom Daraja API (M-Pesa) integration once core auth, cart, and product management are stable.

Week 6 – Database Integration & CRUD Operations

Three in-class activities using a simple `studentdb` schema (register, display, edit/delete), followed by the main submission: a complete **Employee Records Management System** including the four bonus features named in the brief – session-based login, free-text search (parameterised `LIKE` query), responsive layout, and full server-side form validation.

Week 7 — User Authentication and Session Management

A standalone **Student Portal Login System**, deliberately kept independent of ShopMart so the authentication workflow (registration → hashed password → login → session → protected dashboard → logout) could be implemented and verified in isolation.

Week 8 — Responsive Web Design and Mobile-First Development

Three responsive builds, all developed mobile-first and tested at three or more breakpoints in Chrome DevTools:

- **Task 1 — Profile Page:** Flexbox layout, single column on mobile, switching to a row at 600px.
- **Task 2 — Product Showcase:** CSS Grid, `1fr` → `repeat(2,1fr)` → `repeat(4,1fr)` across mobile/tablet/desktop.
- **Extra Assignment — Crumb & Co.:** a full 5-page cake training & supplies site with its own cocoa-brown/gold visual identity, a hamburger nav that collapses below 900px, and markup validated through an HTML parser (not just eyeballed in-browser).

Security Practices Applied Throughout

- **Prepared statements (mysqli)** for every query — no string-concatenated SQL anywhere in the repository.
- **bcrypt password hashing** via `password_hash()` / `password_verify()` — passwords are never stored,

logged, or compared in plain text.

- **Session-based access control** — every protected page re-checks for an active session on every request, not only immediately after login.
- **Server-side validation** on every form, independent of (and in addition to) any client-side JavaScript validation.

Setup & Running Locally

1. Install [XAMPP](#) and start the **Apache** and **MySQL** modules.
2. Clone this repository into your `htdocs` folder:

```
git clone https://github.com/Otyzo/Advanced-web-design-Tasks.git
```

3. Open **phpMyAdmin** (`localhost/phpmyadmin`) and create the relevant database for the week you want to run (e.g. `shopmart` , `studentportal` , `employee_records`).
4. Update the `db.php` / `includes/auth.php` connection credentials in that week's folder if your MySQL username or password differs from the XAMPP default (`root` , no password).
5. Visit the relevant folder in your browser, e.g.:

```
http://localhost/Advanced-web-design-Tasks/shopmart-week5/  
http://localhost/Advanced-web-design-Tasks/week7-student-portal/
```

Each week's folder is runnable on its own — none of them depend on a week outside of the Weeks 1–5 ShopMart

sequence, which is intentionally cumulative.

Documentation

The full practical logbook — covering objectives, screenshots, reflections, and weekly reflection questions for every week — is in [Documentation/Consolidated Logbook Week1-8.docx](#).

Author

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